

QUESTIONS

1. Pearl Company paid dividend 30 TL per share last year. The stocks currently sell 500 TL per share. It is estimated 8% growth per year. What is the cost of equity?

Solution: $D_1 = D_0 * (1+g) = 30 * (1+0,08) = 32,4$

$R_e = (D_1 / P_0) + g = 32,4 / 500 + 0,08 = 14,48 \%$

2. Calculate the aftertax cost of debt under each of the following conditions:

<u>Yield</u>	<u>Corporate Tax Rate</u>
<u>a. 8.0%</u>	<u>18%</u>
<u>b. 12.0%</u>	<u>34%</u>
<u>c. 10.6%</u>	<u>15%</u>

Solution: $K_d = \text{Yield} (1 - T)$

	<u>Yield</u>	<u>(1 - T)</u>	<u>Yield (1 - T)</u>
a.	8.0%	(1 - .18)	6.56%
b.	12.0%	(1 - .34)	7.92%
c.	10.6%	(1 - .15)	9.01%

3. The treasurer of Riley Coal Co. is asked to compute the cost of fixed income securities for her corporation. Even before making the calculations, she assumes the aftertax cost of debt is at least 3 percent less than that for preferred stock. Based on the following facts, is she correct?

Debt can be issued at a yield of 11.0 percent, and the corporate tax rate is 21 percent. Preferred stock will be priced at \$60 and pay a dividend of \$6.40. The flotation cost on the preferred stock is \$6.

Solution: Riley Coal Inc.

Aftertax cost of debt:

$$\begin{aligned} K_d &= \text{Yield} (1 - T) \\ &= 11.0\% (1 - .21) = 11.0\% (.79) = 8.69\% \end{aligned}$$

Aftertax cost of preferred stock:

$$K_p = \frac{D_p}{P_p - F} = \frac{\$6.40}{\$60 - \$6} = \frac{\$6.40}{\$54} = 11.85\%$$

Yes, the treasurer is correct. The difference is 3.16% (8.69% versus 11.85%).

4. Given the following information, calculate the weighted average cost of capital for Digital Processing Inc. Line up the calculations in the order shown in the Table.

Percent of capital structure:

Preferred stock	20%
Common equity	40
Debt	40

Additional information:

Corporate tax rate	25%
Dividend, preferred	\$8.50
Dividend, expected common	\$2.50
Dividend, preferred	\$105.00
Growth rate	7%
Bond yield	9.5
Flotation cost, preferred	\$3.60
Price, common	
.....	\$75.00

Solution: Digital Processing Inc.

$$\begin{aligned} K_d &= \text{Yield} (1 - T) \\ &= 9.5\% (1 - .24) \\ &= 9.5\% (.76) \\ &= 6.27 \end{aligned}$$

$$\begin{aligned} K_p &= D_p / (P_p - F) \\ &= \$8.50 / (\$105 - 3.60) = \$8.50 / \$101.40 = 8.38\% \end{aligned}$$

$$\begin{aligned} K_e &= (D_1 / P_0) + g \\ &= (\$2.50 / \$75) + 7\% = 3.33\% + 7\% = 10.33\% \end{aligned}$$

	<u>Cost (aftertax)</u>	<u>Weights</u>	<u>Weighted Cost</u>
Debt (K_d).....	7.22%	40%	2.89%
Preferred stock (K_p).....	8.38	20	1.68
Common equity (K_e) (retained earnings).....	10.33	40	4.13
Weighted average cost of capital (K_a)....			8.70%

5. Speedy Delivery Systems can buy a piece of equipment that is anticipated to provide an 11 percent return and can be financed at 6 percent with debt. Later in the year, the firm turns down an opportunity to buy a new machine that would yield a 9 percent return but would cost 15 percent to finance through common equity. Assume debt and common equity each represent 50 percent of the firm's capital structure.

a. Compute the weighted average cost of capital.

b. Which project(s) should be accepted?

6. Telecom Systems can issue debt yielding 9 percent. The company is in a 30 percent bracket. What is its aftertax cost of debt?

7. Medco Corporation can sell preferred stock for \$90 with an estimated flotation cost of \$2. It is anticipated the preferred stock will pay \$8 per share in dividends.

a. Compute the cost of preferred stock for Medco Corp.

b. Do we need to make a tax adjustment for the issuing firm?

8. Global Technology's capital structure is as follows:

Debt..... 35%

Preferred stock 15

Common equity..... 50

The aftertax cost of debt is 6.5 percent; the cost of preferred stock is 10 percent; and the cost of common equity (in the form of retained earnings) is 13.5 percent. Calculate Global Technology's weighted average cost of capital.

9.

<u>Input areas</u>	
<u>Growth rate</u>	7%
<u>Last dividend</u>	\$1.76
<u>Stock price</u>	\$52.99

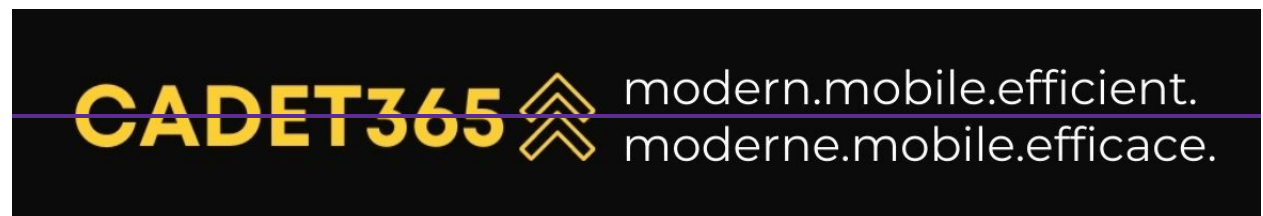
According to the input areas, what is the cost of common equity?

10. Murray Motor Company wants you to calculate its cost of common stock. During the next 12 months, the company expects to pay dividends (D_1) of \$2.50 per share, and the current price of its common stock is \$50 per share. The expected growth rate is 8 percent.

a. Compute the cost of retained earnings (K_e).

b. If a \$3 flotation cost is involved, compute the cost of new common stock (K_n).

Cadet365 - Onboarding with a Personal Computer (PC) and a Smartphone



Step by step instructions

Please note: It is **highly recommended** you have Chrome or Edge installed and use it for the onboarding process. Please note your smartphone must be passcode protected to move forward.

1. On a computer, using Chrome or Edge browser, navigate to office.com

2. Sign into your account:

- Staff: Enter your CCO account credentials.
- Cadets: Enter the email and temporary password that was provided to you.

3. Read the **DND - Cadets Terms of Use** (click on the **CJCR-Terms of Use** to expand it). Scroll down and click "**Accept**".

4. Follow these on screen instructions:

- On the **More information required** page, click **"Next"** - yes
- On the Keep your account secure, Microsoft Authenticator, **Start by getting the app** page, click **"Next"**
- On the Keep your account secure, Microsoft Authenticator, **Set up your account** page, click **"Next"**
- You will then get to the page displaying the QR code

5. On your Smartphone go to your applications store and download the **Microsoft Authenticator** app. Click on the icon below or search for **"Microsoft Authenticator"** on the App Store (iOS) or Google Play (Android) to download the app (you will leave the Government of Canada's website).

6. Open the **Microsoft Authenticator** app.

- Click the three dots on the top right (or the + plus symbol on some devices);
- Select **Add account**;
- Select **Work or school account**;
- Click **Scan a QR code**.
- A camera overlay will pop up

7. **Scan the QR code** on your computer with your smartphone.

- Close the app on your phone.
- On your computer, click **"Next"**. Your computer the screen will go to **Let's try it out**.

8. You will be pinged on your phone to **Approve the sign-in**.

- You will see a small notification on your phone. Tap on it to reveal more options.

- The notification expands into a pop-up window. Approve the sign-in request by tapping on **"Approve"**.
- Since App Lock is enabled, you will be prompted to unlock Microsoft Authenticator using the PIN, fingerprint or other unlock mechanism you have set for your smartphone.
- Close the **Microsoft Authenticator** app.
- Your computer screen will automatically update to show **Notification approved**. Click **"Next"**.

9. Follow the on screen instructions to add a second authentication method e.g. phone number. To do this, enter your phone number, select Canada (+1), select **"Text me a code"** and then **"Next"**. You will receive a text message with a code on your phone. Enter the code received by text message then select **"Next"**. Then **"Next"** again and then **"Done"**.

- From now on, if you do not have access to the authentication tool to log into Cadet365, you can select the **"Sign in another way"** option to display and choose the **"Phone"** option.
- On your phone's Authenticator app, you will see all the accounts you have to authenticate with your device. If this is your first time doing this, you will only see your Cadet365 account. Every time you log into Cadet365 on your PC, you will receive a push notification on your device to approve your login.

10. The screen will update to **"Stay signed in?"**, selected **"Yes"** (unless other users access your computer). **The Cadet365 platform will load. Congratulations - you have successfully onboarded!**